THE INDEPENDENT INSTITUTE OF EDUCATION

EMERIS VEGA SCHOOL

GADE7322: Game Development 3B

Portfolio of Evidence (PoE): Group Project

Project Links, References and Acknowledgements

Procedural Tower Defence (ProceduralGenGade3B)

Group members:

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Module: GADE7322

Year: 2026

GitHub repository:

https://github.com/harrietaleck/ProceduralGenGade3B

Referencing style: Harvard Anglia (The Independent Institute of Education, 2025)

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Project repository

All source code, Unreal Engine content and project files for the group game are stored in one GitHub repository (Manda and Mtabane, 2026). The branches below hold each stage of the Portfolio of Evidence.

Table 1: Repository links and project details

Item	Link or detail
Repository	https://github.com/harrietaleck/ProceduralGenGade3B
Part 1 (procedural terrain and core loop)	https://github.com/harrietaleck/ProceduralGenGade3B/tree/part1
Part 1 user interface work	https://github.com/harrietaleck/ProceduralGenGade3B/tree/part1UI
Part 2 (enemy types, defenders, procedural waves)	https://github.com/harrietaleck/ProceduralGenGade3B/tree/part2
Engine	Unreal Engine 5.8, C++ with UMG widgets
Plugins	ProceduralMeshComponent, Modeling Tools Editor Mode
How to run	Check out the branch, run git lfs pull, open ProceduralGenGade3B.uproject and press Play on the TowerDefense map

Acknowledgements

Third-party assets

- Low Poly Nature: Essentials by Vertex Rage Studio (2024) supplies the trees, rocks, plants and props that the terrain scatters across the map (Content/VRS_LowPolyNatureEssentials). It was used under the Fab / Unreal Engine Marketplace licence and was not modified.
- User interface screen artwork in Content/UI/SourceArt (start, settings, defenders, upgrades, game over and victory screens): [confirm creator and tool or source].

Engine, tools and documentation

- Unreal Engine 5.8 and its documentation (Epic Games, 2025a), in particular UProceduralMeshComponent for the runtime terrain mesh (Epic Games, 2025b) and UUserWidget for the HUD and menu screens (Epic Games, 2025c).
- Git, GitHub and GitHub Desktop for version control and collaboration.

Algorithms and research

- Enemy rerouting uses the A* search algorithm (Hart, Nilsson and Raphael, 1968) over the procedural path network.
- Enemy paths and rerouted routes are smoothed with Chaikin's corner-cutting algorithm (Chaikin, 1974).
- Procedural generation design was informed by Farrokhi Maleki and Zhao (2024). The adaptive wave director was informed by research on dynamic difficulty adjustment (Guo, Thawonmas and Ren, 2024; Mortazavi, Moradi and Vahabie, 2024) and adaptive wave generation in tower defence (Hind and Harvey, 2024).
- Assessment requirements come from the GADE7322 Part 1 and Part 2 briefs and rubrics (The Independent Institute of Education, 2026a; 2026b).

Use of generative AI

The Cursor AI coding agent (Anysphere, 2026) was used during development to help write and debug C++ gameplay code, rewrite code comments in plain language, and draft planning and report documents. Commits made with this tool are marked in the GitHub history with a "Co-authored-by: Cursor" line. All AI-assisted output was reviewed, play-tested and adjusted by the group, and the group made the design decisions, tested the game and is responsible for the final submission. [Group members: confirm or adjust this statement so it matches how each of you used AI tools.]

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